

IPsec Security Assessment Report

AES128_SHA256_DH14_NOPFS_ICMP_rep01 | EXECUTIVE | 2026-09-27T16:55:09.050Z

Assessment scores

Security 54/100 | Observed risk 10/100 | Evidence coverage 60% (PARTIAL)

Evidence-adjusted score: 54%



Evidence coverage: 60%



Executive summary

The IPsec capture (40 packets, 6.8 KB) shows an IKEv2 exchange using AES-128-CBC and HMAC-SHA2-256-128 over DH Group 14. The overall score is 54 (Weak), with 60% evidence coverage. One low-severity observed risk (F1) flags the use of CBC mode, which is susceptible to padding oracle attacks if timing side-channels exist. Two low-severity evidence gaps (F2, F3) indicate that Perfect Forward Secrecy and anti-replay window settings could not be assessed from the capture. Traffic features suggest a VoIP/audio pattern (60% relative pattern score, derived from observed data), but encrypted payload contents are not visible. No confirmed vulnerabilities or compliance claims are made.

Security and risk

Evidence-adjusted security score: 54/100 (Weak).

Observed configuration risk score: 10/100. This is a rule-based penalty, not the probability of an attack.

Evidence coverage: 60% (PARTIAL). Unknown controls receive no security credit.

Traffic and metadata inference

Inferred workload: VoIP / Audio Call.

Traffic pattern match: 60%. This is a relative heuristic score, not calibrated model confidence.

Wire metadata: 6 IKE and 40 ESP frames; NAT traversal Not detected.

Application identity and encrypted contents cannot be confirmed from packet metadata.

AI confidence

AI confidence score: 93%.

Mean predicted-class probability across 4 available cryptographic inference models; this is not measured accuracy.

Threat matrix and remediation

Severity	Control / threat	Observed	Penalty	Action
Low	Symmetric Encryption Cipher: Cipher Block Chaining Mode Padding Attacks	AES-CBC	10	Prefer Authenticated Encryption with Associated Data (AEAD) modes like AES-GCM or ChaCha20-Poly1305.

Evidence gaps

Perfect Forward Secrecy (PFS): Not observed in capture. Capture CREATE_CHILD_SA exchanges or provide an explicit Child SA configuration.

Anti-Replay Window Protection: Not observed in capture. Provide Child SA configuration or negotiated ESN evidence.